

Model-based reinforcement learning for controlling piecewise deterministic semi-Markov decision processes that are partially observable

Orlane Rossini

Supervised by Alice Cleynen, Benoîte de Saporta and Régis Sabbadin

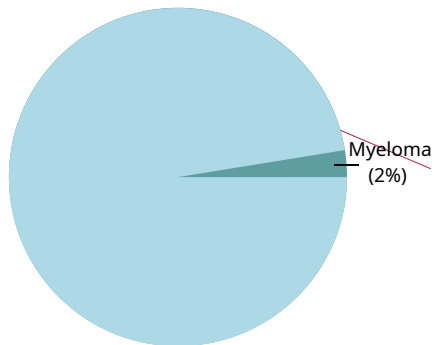
November, 28 2025



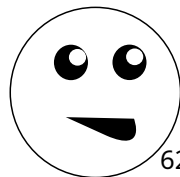
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Medical context: Multiple myeloma



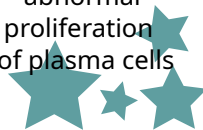
Distribution of cancers in France (2018)



62%

5-year survival

abnormal
proliferation
of plasma cells



Medical context: Multiple myeloma

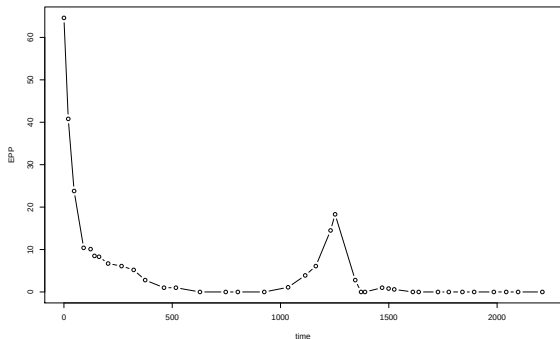


FIGURE: Example of patient data [ALa17]

- Patients who have had **cancer** benefit from **regular follow-up**;
- The concentration of clonal immunoglobulin is measured over time;
- The doctor has to make new **decisions** at each visit:
 1. delay for the next visit (15, 30 or 60 days),
 2. treatment (chemotherapy or not).

Medical context: Multiple myeloma

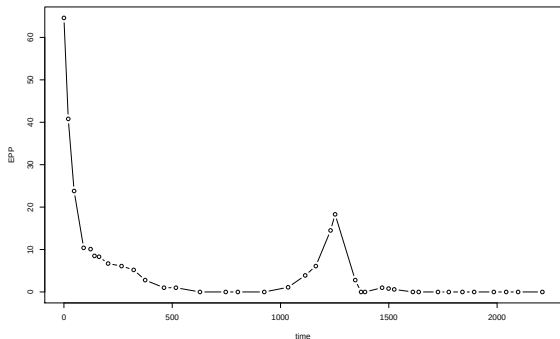
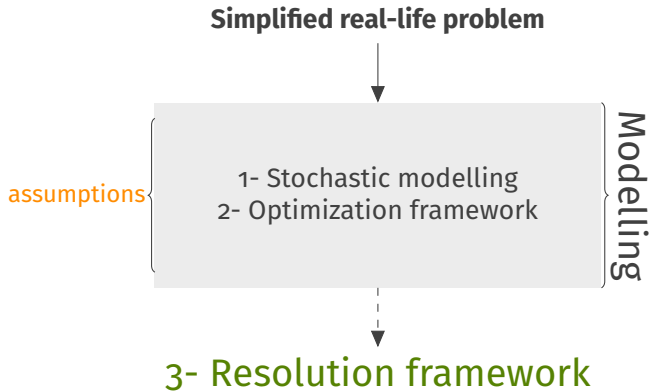


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⇒ **Optimising decision-making to ensure the patient's quality of life**

Work approach

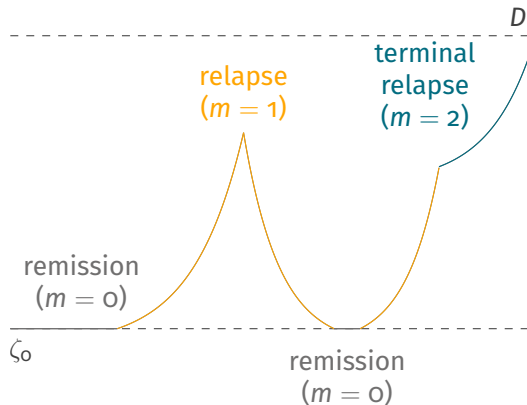


Contents

- ▶ MODELLING THE PROBLEM
- ▶ CONTINUOUS TIME AND STATE SPACE
- ▶ PARTIAL OBSERVATION
- ▶ PARTIALLY KNOWN DYNAMICS
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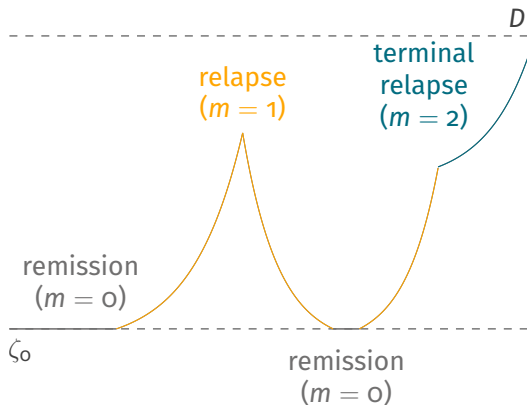
Controlled PDMP

We switch **randomly** from one **deterministic** regime to another.



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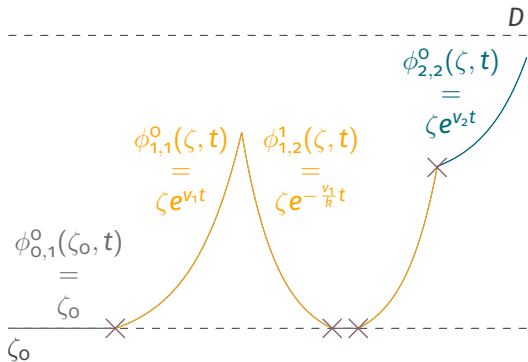


Let $x = (m, \ell, k, \zeta, u)$ the patient's state:

- m the patient's condition;
- ℓ the current treatment;
- k the number of treatments;
- ζ the biomarker;
- u the time since the last jump.

Local Characteristics of a PDMP

A PDMP is defined by three local characteristics.



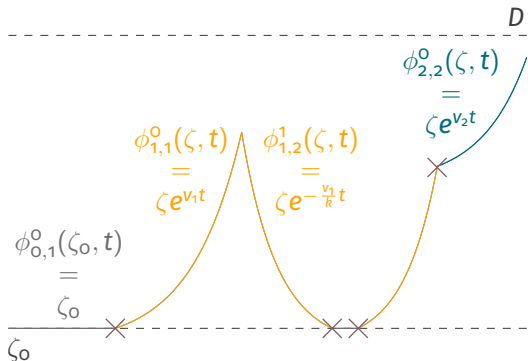
FLOW

Description of the deterministic part of the process.

$$\Phi(x, t) = (m, k, \ell, \phi_{m,k}^\ell(\zeta, t), u + t)$$

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JUMP INTENSITY

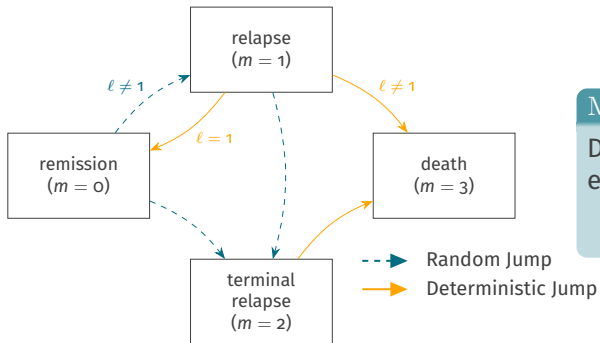
Description of the process jump mechanisms.

- Boundary jump (deterministic)
- Random jump

$$\lambda(x) = \begin{cases} (\mu_{m,\ell,k}^1 + \mu_{m,\ell,k}^2)(u) & \text{if } m = 0, \\ \mu_{m,\ell,k}^2(\zeta) & \text{if } m = 1, \\ 0 & \text{otherwise.} \end{cases}$$

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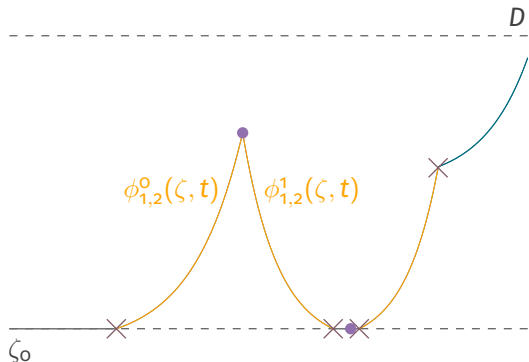


MARKOV KERNEL

Description of the state of the process after each jump.

$$\mathbb{P}(X_t \in A | X_{t-} = x) = Q(x, A)$$

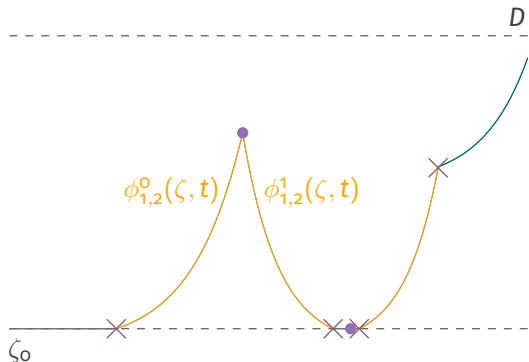
Impulse control for PDMP



Choosing a new starting point :

- date for the next impulse;
- point from which to restart the process.

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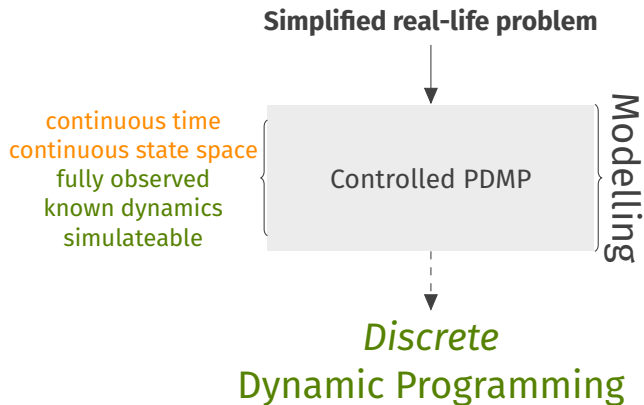
Restrictions:

- the delay between consecutive impulses is in a finite set;
- only change the current treatment ℓ .

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A fully observed and known process



Optimization framework of impulse control for PDMP

Identify an ϵ -optimal strategy $\mathcal{S}^\epsilon = (\tau_n, \mathcal{L}_n)_{n \geq 1}$

$$\underbrace{\mathcal{J}(\mathcal{S}, x)}_{\text{Expected cost of strategy } \mathcal{S}} = \mathbb{E}_x^{\mathcal{S}} \left[\int_0^H e^{-\gamma t} \underbrace{c_R(X_t)}_{\text{running cost}} dt + \sum_{n=1}^H \underbrace{c_I}_{\text{impulse cost}}(X_{\tau_n}, X_{\tau_n}^+) \right],$$

with

- $c_R(X_t) = C_T \mathbb{1}_{\ell=1} + C_D \mathbb{1}_{m=3},$
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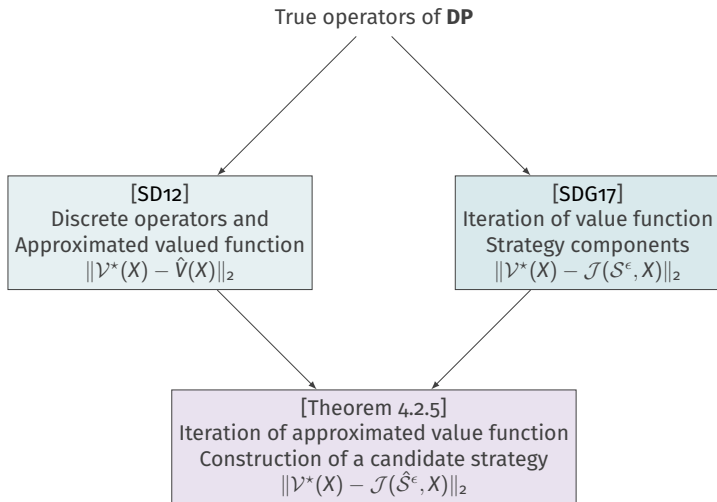
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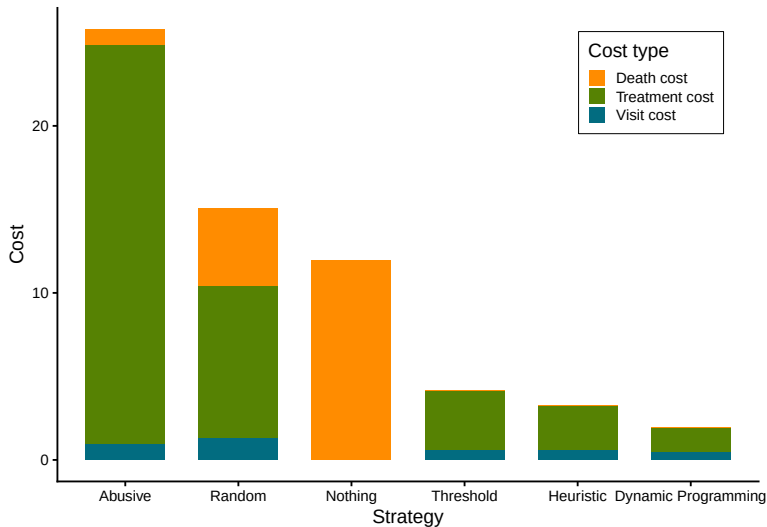
$$\mathcal{V}^*(x) = \inf_{\mathcal{S} \in \mathcal{S}} \mathcal{J}(\mathcal{S}, x)$$

$$\mathcal{V}^*(x) \leq \mathcal{J}(\mathcal{S}^\epsilon, x) + \epsilon$$

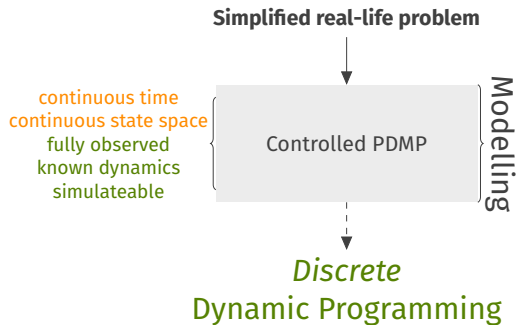
Candidate strategy and optimality



Preliminary results



Summary of contributions

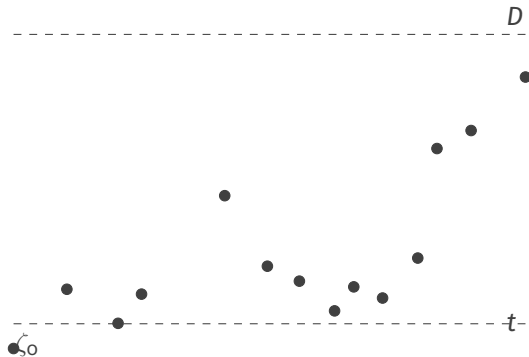


- + **Numerical method** yielding an explicit control strategy
- + **Theoretical guarantees** ensuring the computed strategy is ϵ -optimal
- **Limited applicability** in more realistic situations

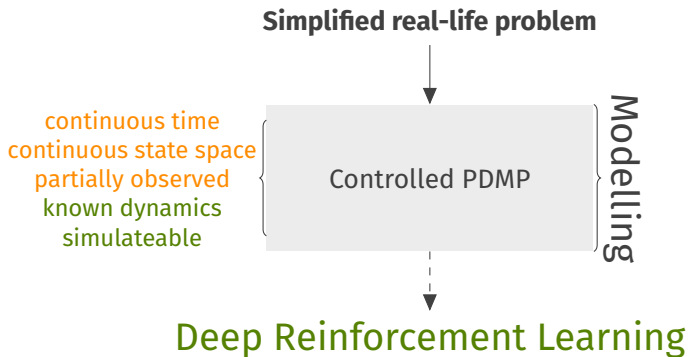
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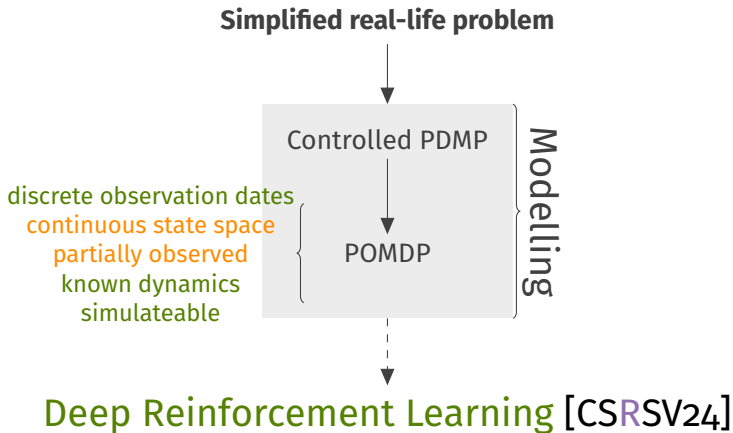
The problem of partial observation



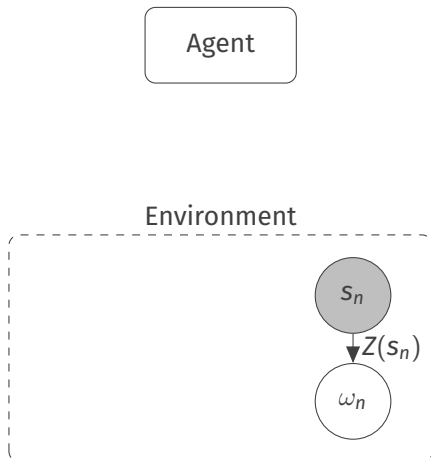
A partially observed process



A partially observed process



Characteristics of a POMDP

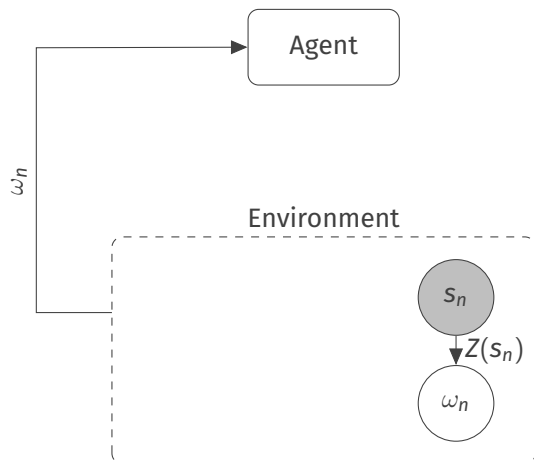


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 $s = (m, k, \zeta, u) \in \mathcal{S}$;
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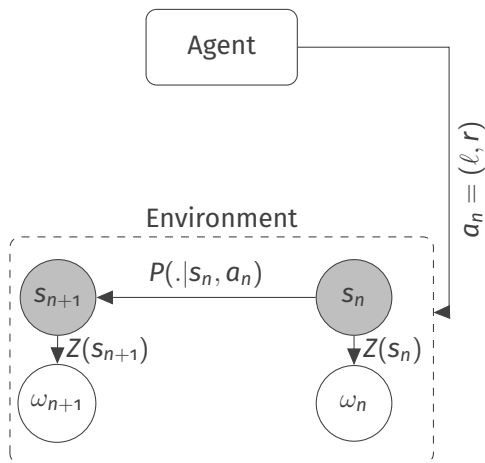


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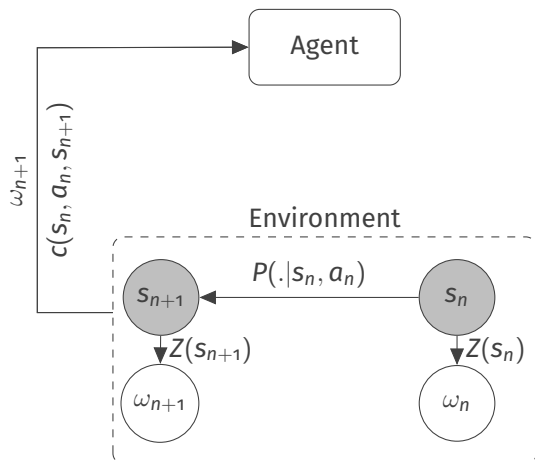
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Optimization framework of POMDP

Identify an optimal policy $\pi^\star$

$$\underbrace{c(s, a, s')}_{\text{Cost function}} = \underbrace{C_V}_{\text{visit cost}} + \underbrace{C_D \times \mathbb{1}_{m'=3}}_{\text{death cost}} + \underbrace{C_T \times r \times \mathbb{1}_{\ell=a}}_{\text{treatment cost}}$$

Optimization framework of POMDP

Identify an optimal policy π^*

$$\underbrace{V(\pi, s)}_{\text{Optimization criterion}} = \underbrace{\mathbb{E}_s^\pi \left[\sum_{n=0}^{H-1} c(S_n, A_n, S_{n+1}) \right]}_{\text{Expected total cost as a result of the policy } \pi}$$

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Optimization framework of POMDP

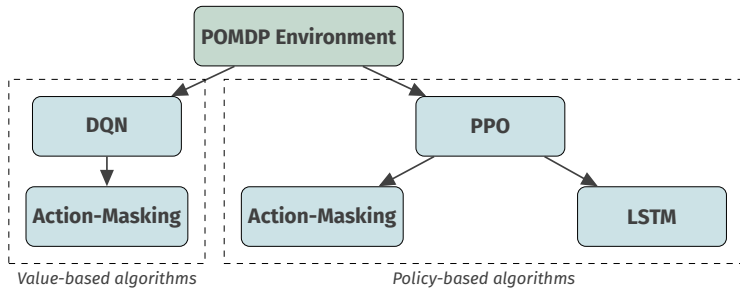
Identify an optimal policy π^*

In reality, we do not observe state space!

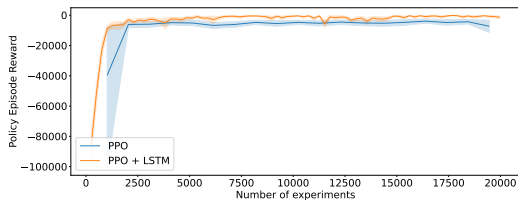
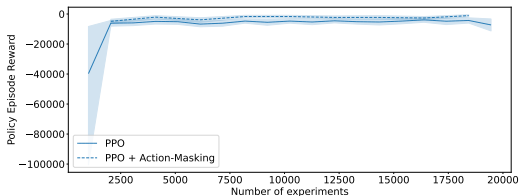
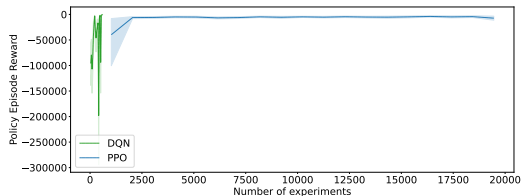
Let $h_n = (\omega_0, a_0, \omega_1, a_1, \dots, \omega_n)$ be the history

$$\underbrace{V^*(h)}_{\text{Value function}} = \underbrace{\min_{\pi \in \Pi} V(\pi, h)}_{\text{Minimisation across policy space.}}$$

Solving with Deep Reinforcement Learning



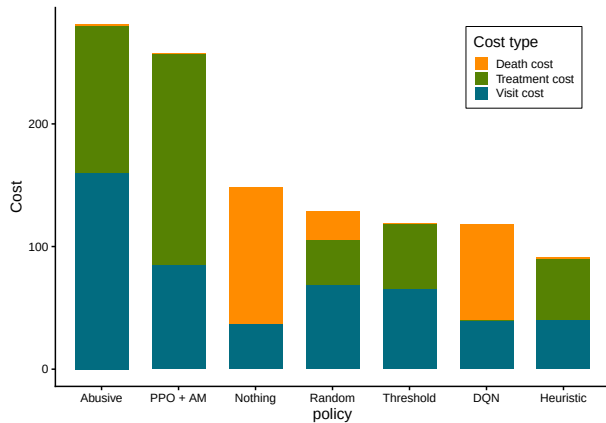
Benchmark Results



⇒ **DQN** and **PPO with Action-Masking** outperformed all baseline algorithms.

Results

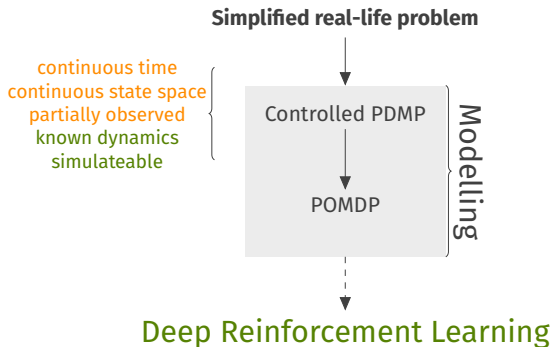
Policy evaluation performance on 5 000 Monte-Carlo simulations



Obtained policies

Policy	Survival rate	Visit delay	Number of treatments	Time under treatment
Abusive	99.76%	15	1	6 years
Nothing	62.76%	60	0	0
Random	92.16%	34.82	4.34	1 year and 1 month
Threshold	99.99%	40.42	4.97	2 years and 7 months
Heuristic	99.20%	60	1.01	1 year and 4 months
PPO with AM	99.83%	28.40	26.34	4 years and 6 months
DQN	73.91%	59.07	0.80	47 days

Summary of contribution



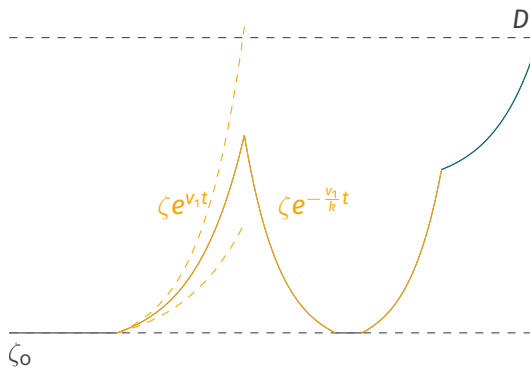
- + **Alternative resolution method** for POMDP derived from PDMP
- **Usage of LSTM** for partial observation
- **Limited interpretation** of policies
- **Required** full PDMP knowledge

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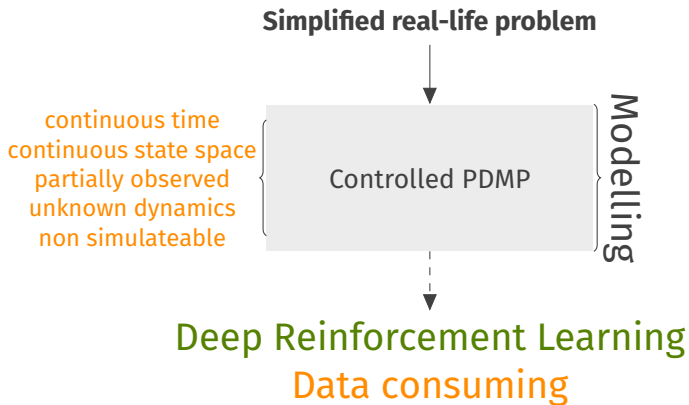
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The problem of partially known dynamics

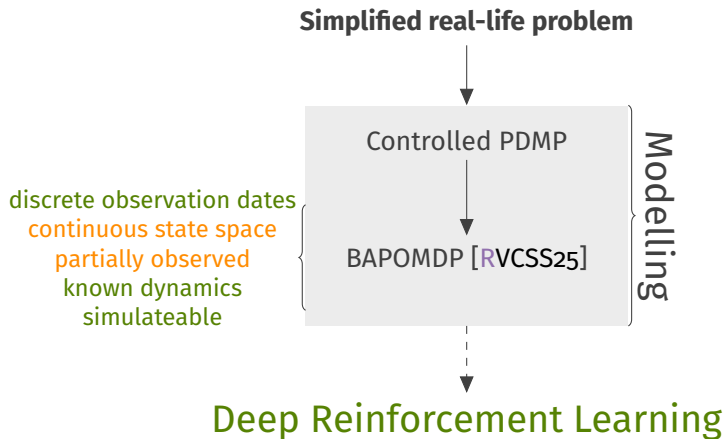
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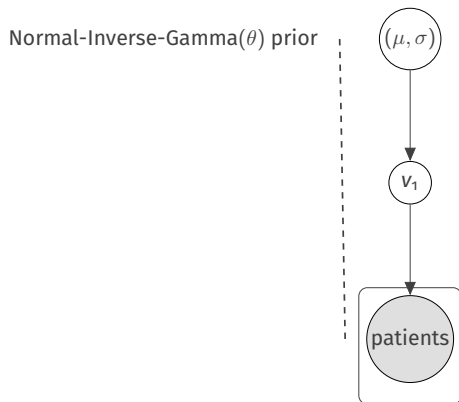
A partially observed and known process



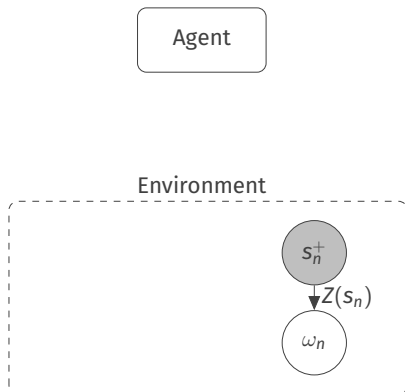
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Handle uncertainty with Bayesian framework



Characteristics of a BAPOMDP

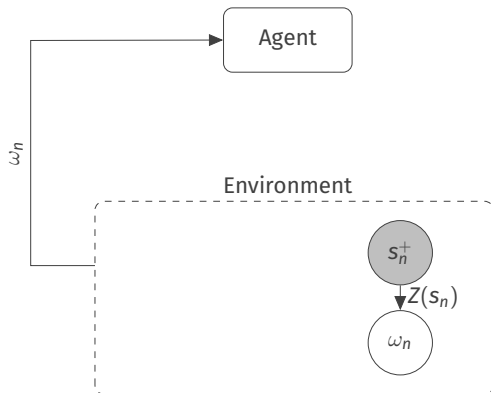


BAPOMDP DEFINITION

Un BAPOMDP se définit par un tuple $(\mathbb{S}^+, \mathbb{A}, P^+, \Omega, Z, c)$.

- **Space of hyperstate** $\mathbb{S}^+ = \mathbb{S} \times \Theta$;
- Actions $a = (\ell, r) \in \mathbb{A}$;
- **Transition function** $P^+(s', \theta' | s, a, \theta)$;
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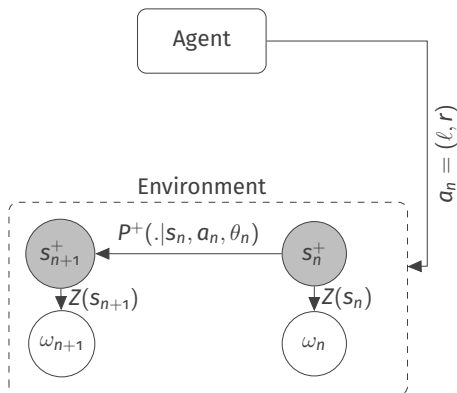


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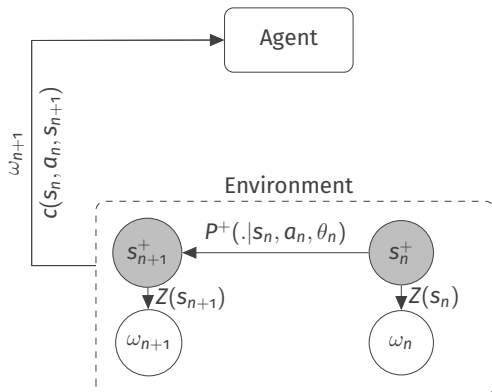
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$$P^+((s', \theta') \in B_E \times B_\Theta \mid (s, \theta), a)$$

$$= \int_{B_E} \mathbf{1}_{B_\Theta}(\mathcal{U}(\theta, s, a, s')) \times P(ds' \mid s, a, \theta).$$

Characteristics of a BAPOMDP



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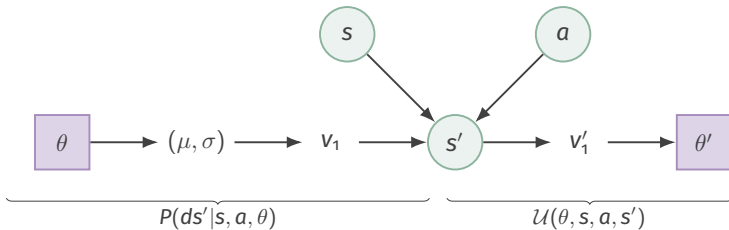
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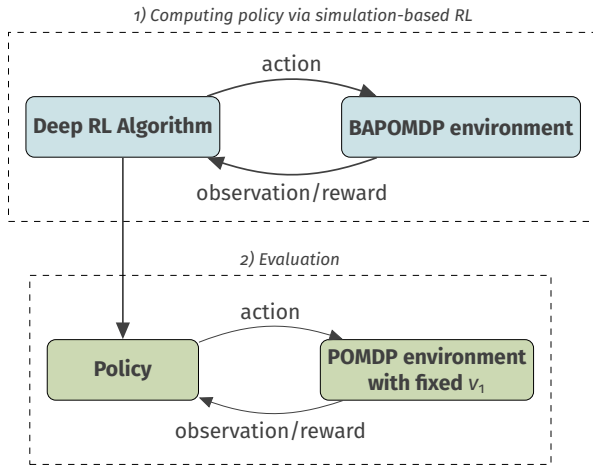
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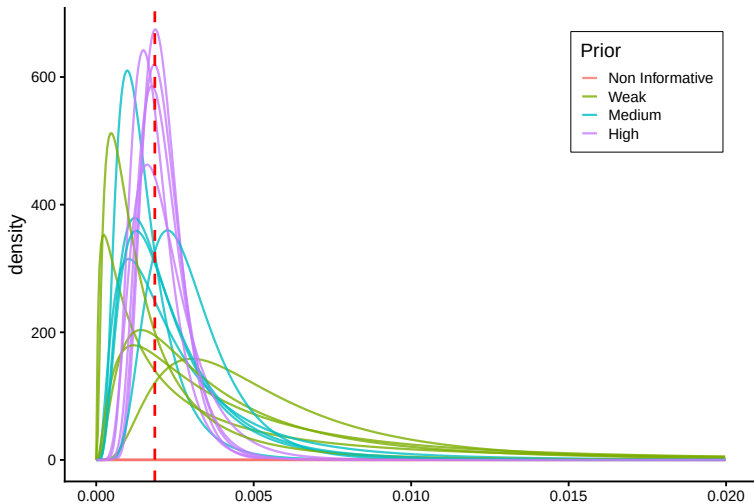
Generate transition from prior



Evaluate BAPOMDP framework

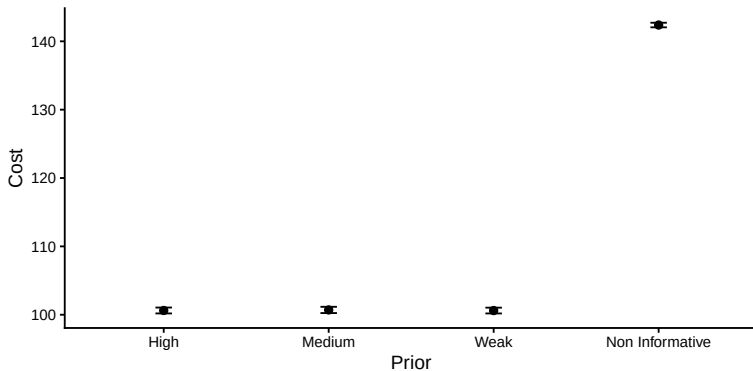


Different prior



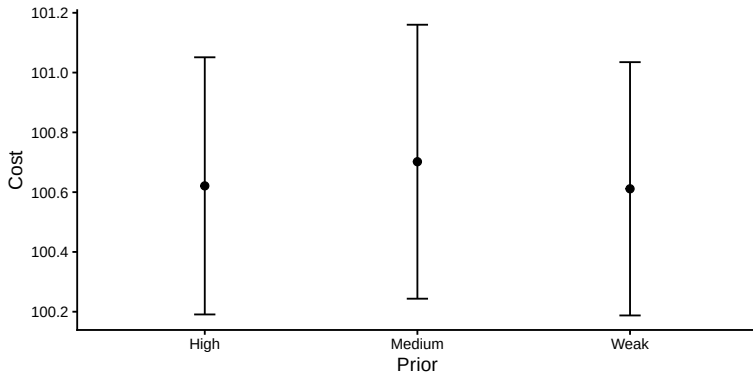
Preliminary results

Policy evaluation robustness on 5 000 Monte-Carlo simulations for DQN with Action-Masking

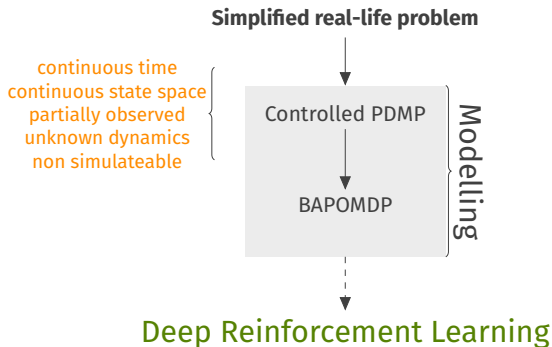


Preliminary results

Policy evaluation robustness on 5 000 Monte-Carlo simulations for DQN with Action-Masking



Summary of contribution



- + **Novel equivalence framework** to represent PDMPs with unknown parameters as BAPOMDPs yielding an explicit control strategy
- + **Deep reinforcement learning** can be leveraged to solve the resulting BAPOMDP formulation
- **No estimates** of unknown parameters

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General Conclusion: How to solve a controlled PDMP ?

Assumptions	Discrete DP	POMDP + DRL	BAPOMDP + DRL
Continuous time	✓	✓	✓
Continuous state space	✓	✓	✓
Large state space	✗	✓	✓
Partial observation	✗	✓	✓
Unknown parameters	✗	✗	✓
Theoretical guarantees	✓	✗	✗

A specific problem that can be extended to many others !

Stochastic modelling

- Optimization framework: cost function calibration
- Modeling more complex models
- Extension of this resolution framework to other Markov models (HsMM)

Artificial intelligence

- New benchmark to evaluate Deep RL algorithms
- Continuous-time modeling framework compatible with RL
- For BAPOMDP: real-environment interaction to update parameters and policy online

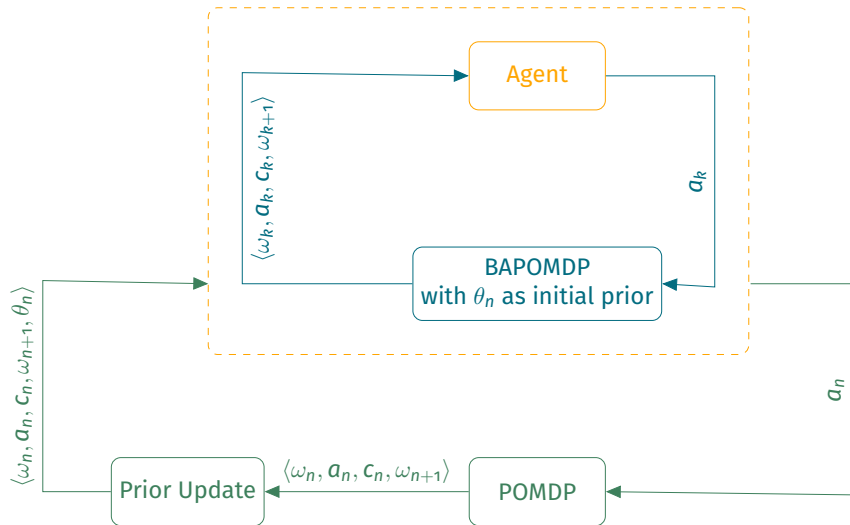
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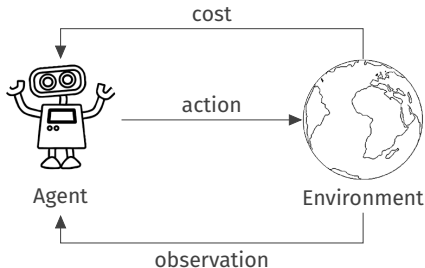
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Future work



Reinforcement Learning



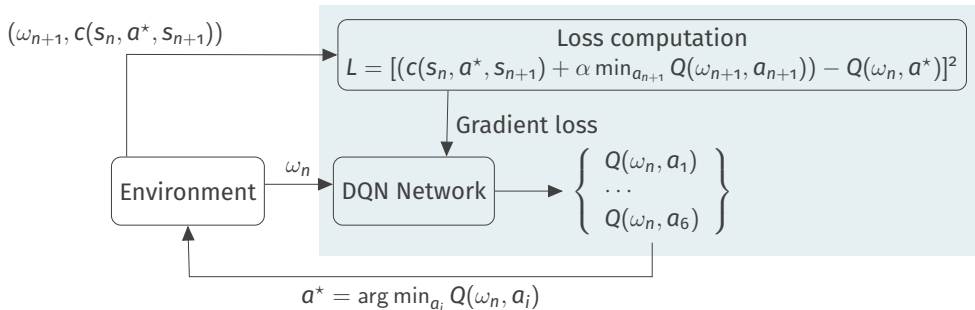
The optimal policy is obtained from the experiments $\langle \omega, a, \omega', c \rangle$, generated from P^+ transition function

$$\underbrace{Q^\pi(s, a)}_{\text{Q value}} = \underbrace{\mathbb{E}^\pi \left[\sum_{n=0}^{H-1} c(S_{n-1}, A_n, S_n) \mid s, a = (\ell, r) \right]}_{\text{Value of an action in a state according to the policy } \pi}$$

$$\underbrace{Q^*(s, a)}_{\text{Q function}} = \min_{\pi \in \Pi} Q^\pi(s, a)$$

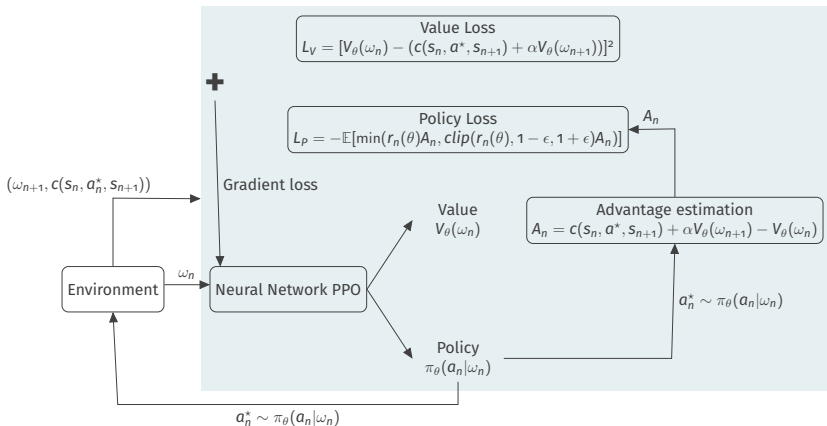
$$\underbrace{A(s, a)}_{\text{Advantage function}} = \underbrace{Q(s, a) - V(s)}_{\text{Extra cost obtained by the agent by taking the action}}$$

Algorithm example: DQN¹



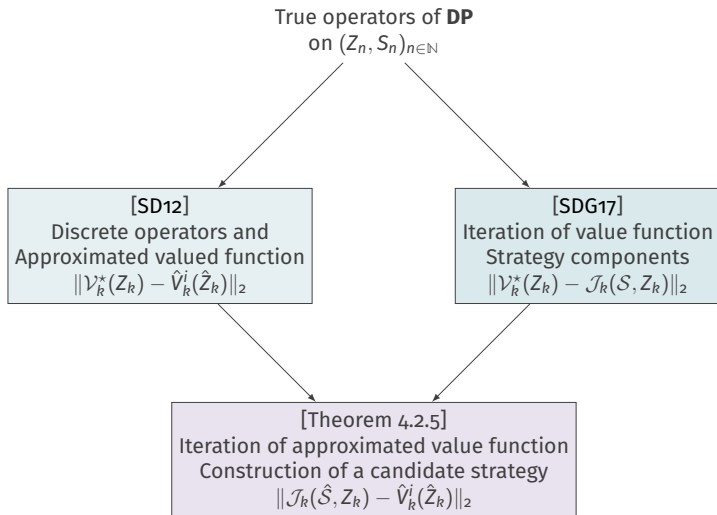
¹Deep Q-Network

Algorithm example: PPO²



²Proximal policy optimization

Candidate strategy and optimality



Framework comparison

BAPOMDP VS POMDP WITH A FIXED PRIOR

Policy evaluation robustness on 5 000 Monte-Carlo simulations for DQN with Action-Masking

